Christy Gonzola (Undergraduate Student - Molecular & Cell Biology)

"I can do everything you can. It just takes me a little bit longer."

Background

Age: 20

Occupation: Undergraduate student, second year

School: Molecular & Cell Biology, University of California, Berkeley

Technology level: Familiar with adaptive technologies

Main Points

- Has low vision and uses various assistive technologies
- Spends a lot of time getting through school work
- Is very social and loves to spend time with friends
- Wants to reach out and help other students with similar challenges

Goals

- Efficiently select and use various assistive technologies
- Help out other students with visual impairment and other disabilities
- Graduate
- Spend time with friends
- Build the skills necessary to become a cell biologist, but by doing only the work that is necessary

Frustrations & Pain Points

- Few assistive technology-enabled computers on campus
- Inability to scan through documents for relevant information; must listen to the screen reader read the whole document
- Problems keeping track of what she has to have done for school; wishes the LMS could remind her when assignments are due
- Slowness of the LMS
- Hard to scan for new Announcements in the LMS
- Every professor has a syllabus/office hours info in different places
- Too many emails letting her know about documents which have been added to the LMS
- "It would help me if all my coursework were in one place---I have to open so many tabs." (in the LMS)

Scenarios

- Listening to readings with a screen reader
- Taking notes in class
- Completing group lab assignments

Description

Christy is an undergraduate student with very low vision, and lives in the student residence on campus. She likes it because it's a short walk to classes and has a lot of helpful features for people with special needs, "it's got ramps and hand rails everywhere!" She feels pretty comfortable walking to classes and other buildings by herself. She's interested in becoming a Cell Biologist and is taking 5 classes this semester, most of which are large lectures with labs like BIO 1A. Her French class is only ~20 other students so it's quite different from most of her science classes. In her science classes she feels like she needs to learn too much material in the time given. For example, Organic Chemistry is too much.

Most of Christy's classes have sites in bSpace (the Sakai Learning Management System), but some use them more than others. It's always a struggle to keep track of everything she has to do with her courses, study sessions, etc. She says, "it would help me if all my coursework were in one place -- I have to open so many tabs." Christy often finds herself creating "to do" lists but infrequently follows through with checking things off the list. Somehow she manages to stay on top of things. She often wishes that her bSpace course sites could help her "remember" when assignments are due. She'd also love to be able to have some sort of electronic Post-it notes on the computer to collect links and remind her what to read or get done tomorrow. Whenever she has extra time she organizes her typed-up notes on her laptop.

Christy's low vision makes it difficult to use the mouse when she's using a computer. She uses the keyboard heavily and often she uses screen magnifier to see portions of the screen, but she finds it's quicker to use text-to-speech support in conjunction with the keyboard. Christy has a laptop because there is only one computer lab on campus for students with special needs that has assistive technology-enabled computers and it isn't always convenient for Christy to go there, especially if she's working on projects with other students. She consciously tries to give files simple and meaningful names and text labels, so she can access them easily using a screen reader. She is also very

aware of the file extensions, because she can't distinguish the file types very well by the little icons.

A good day for Christy is a day with only 2 classes, where she gets out at 2:00pm and can play games and hang out with friends--one of her favorite things to do is to hang out with friends and be social. She also spends quite a bit of time on the phone talking with friends. Her cell phone doesn't have any special accessibility features built into it, but she figured out how to use the keypads and basic calling and receiving functions, so she doesn't have to strain her eyes all the time trying to see what's on the screen.

On a bad day, Christy has lots of school work to do. She'd really like to be involved in more extracurricular activities, but usually spends so much time trying to get through her school work that she can't find the time. However, she has made time to be a member of an advocacy group on campus for low vision students. The group lobbies to improve campus accessibility for students with visual impairments, to work out the perceived discrimination issues, to help the students with special needs understand what their rights and responsibilities are, and to put them in touch with campus accessibility resources.

Scenarios

- **Listening to readings with a screen reader:** Because of her vision impairment, Christy needs all of her course readings in electronic form. She makes sure she finds electronic copies of her readings, scans them into pdf documents which are then read-out-loud using the screen reader software. One of the limitations she faces is the lack of ability to quickly scan the documents. She has no way of skimming through the text to determine whether the article is useful or not, and has to listen to each article one by one. She finds this very time-consuming.
- **Taking notes in class:** Christy uses her laptop to take notes in class, because if she used pen and paper, it would be difficult for her to read her own handwriting later. For important lectures, she takes a digital voice recorder to class and records the lecture as a backup. Sometimes she gets a volunteer or a friend to take down what's on the board. Seeing the board is very difficult, and it helps her tremendously if the professor speaks what they are writing on the board. Most instructors tend to do this. For her Biology class where the professor often draws cell structure or graphs, she uses a CCTV camera that can zoom in on the board to capture a screenshot of the board. She borrows the camera from the university's assistive technology center.
- **Completing group lab assignments**: When students work together on a lab assignment, they often each complete a section of the weekly assignment alone. Then one person is responsible for putting it all together from each team member's email to them and submitting it.

LMS Use

- Checks her bSpace course sites at least once a day
  - Wants to know when new assignments are posted or she has something due
  - Doesn't care about every little change to the course sites
- Has several classes that have bSpace sites but they don't use them; these tabs are just "noise" and can push her often-used sites' tabs into the "My Active Sites" dropdown where they can be hard to find
  - Took her a while to discover the "My Active Sites" dropdown when she first started using the site with a screen reader. She remembers spending a lot of time looking for a course site, which she eventually found in the dropdown with a friend's help.

Communication

- Uses Outlook email as her main means of digital communication; checks it several times a day
- Uses the Outlook calendar, because it gives her audible alerts
- Prefers the asynchronous mediums such as email or bulletin boards to the synchronous kinds, such as IM, for several reasons:
  - IM applications are usually not very well-supported by screen readers
  - they tend to get very confusing when there are multiple conversations going on at the same time. "it's like listening to 10 conversations at the same time!"

Collaboration

- Doesn't have many group projects for school
- Does work on lab assignments with a group of other students